

SIMATS ENGINEERING
CO3 — ASSESSMENT TOOL 2 — CASE STUDY
COURSE: COMPUTER NETWORKS (CSA0711)

CASE STUDY 1 — VIDEO CONFERENCING QUALITY DEGRADATION

Scenario

A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

1. Is the Problem Primarily Transport-Layer or Application-Layer?

The symptoms — voice distortion, frozen frames, delayed communication — combined with the measured 8% packet loss and 180 ms jitter, point mainly to the transport layer, specifically to how the underlying protocol (or a poor protocol choice) handles loss and timing for real-time media.

The evidence for this diagnosis:

- Jitter (180 ms) is a transport/network-level metric — it measures variation in inter-packet arrival time caused by queuing and route changes, not something the application layer controls directly.
- Packet loss (8%) is extremely high for real-time media; ITU-T guidance treats even 1% loss as noticeable and >5% as severely degrading for voice/video. This magnitude of loss almost always originates below the application — congested links, insufficient QoS prioritization, or a transport protocol reacting badly to loss.
- If the platform is (incorrectly) using a TCP-based transport for the live audio/video streams, TCP's reliable-delivery guarantee would force retransmission and in-order delivery of media packets that are already late — this produces exactly the 'frozen frame, catch-up' behaviour described, because TCP will stall newer frames behind a retransmission of an old, now-useless one.

Application-level buffering strategy is a secondary contributing factor, not the root cause: if the jitter buffer is too small, 180 ms of jitter will cause audible dropouts even over a well-behaved transport; if it is too large, it adds delay on top of network delay. The recommendation therefore treats this as primarily a transport-layer problem with an application-layer buffering component that must be tuned together.

2. Justification Based on Real-Time Multimedia Requirements

Real-time multimedia has fundamentally different delivery requirements from typical data transfer, which is why the transport choice matters so much here:

Requirement | File/Web Transfer (TCP fits) | Live Audio/Video (needs UDP-style transport)

Timeliness | Not critical — data can arrive late | Critical — a late packet is worse than a lost one

Ordering | Must be strict, in-order | Loose — a few out-of-order/lost frames are tolerable

Retransmission | Required for correctness | Counter-productive — retransmitted data usually arrives too late to render

Acceptable loss | Effectively zero | Small, bounded loss (a few %) is acceptable and concealable

Latency budget | Seconds are fine | ITU-T G.114 recommends < 150 ms one-way for acceptable voice quality

Because TCP enforces reliable, in-order delivery, any loss forces it to pause and retransmit, and its congestion-control algorithm reduces the sending rate after loss — both are exactly wrong for real-time media, where a stale audio/video frame is worthless and should be dropped and concealed, not resent. This is why real-time conferencing protocols are built on UDP (carrying RTP/RTCP), which delivers packets as soon as they arrive, tolerates loss, and leaves recovery decisions (skip, conceal, request a key-frame) to the application, which can make a timing-aware choice that TCP cannot.